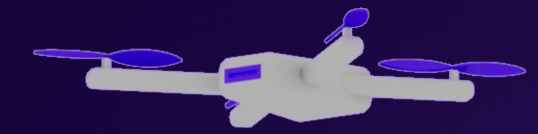
Authentication with .NET MAUI

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david.ortinau@microsoft.com





MODERN CLIENT

David Ortinau

Principal Product Manager

St. Louis, MO – Central Time

Help customers succeed delivering client app experiences across various screens using .NET SDKs for mobile and desktop cross-platform – aka .NET MAUI (and Xamarin).

I like to trail run and have completed multiple distances up to 50 miles.

Grandfather. Doesn't age.

We build .NET MAUI to be the most productive way to develop native apps that perform great on any device that runs Android, iOS, macOS, or Windows from a single codebase.

We build .NET MAUI to be the most productive way to develop native apps that perform great on any device that runs Android, iOS, macOS, or Windows from a single codebase.

.NET MAUI in .NET 6

Unified .NET

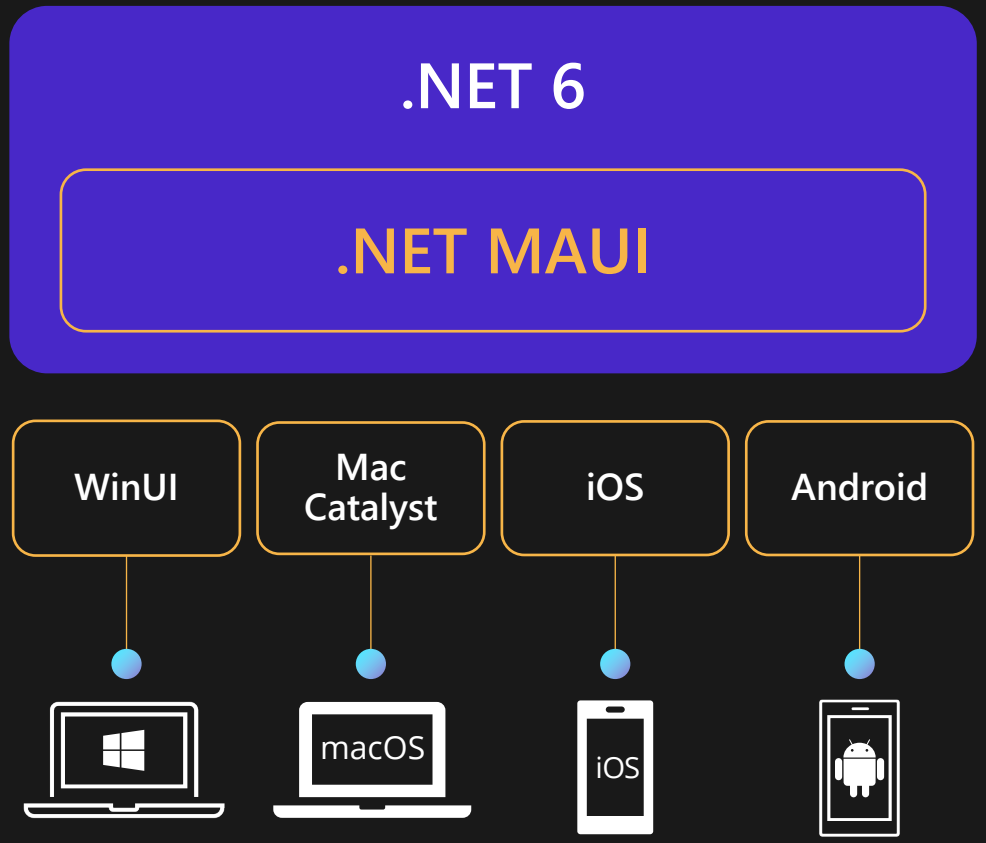
- One Base Class Library
- Type Unification
- CLI Support
- SDK Style Projects
- Modern Support Lifecycle

Mobile, desktop, hybrid

- Android
- iOS
- WinUI
- Mac Catalyst
- Blazor WebView

Productive development

- Single Project
- Microsoft.Extensions
- .NET Hot Reload
- XAML Hot Reload
- Visual Diagnostics





Let’s begin with a

History of Authentication

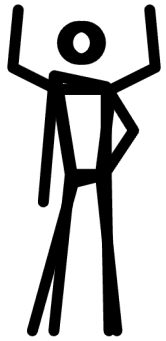
Meh…ask someone else

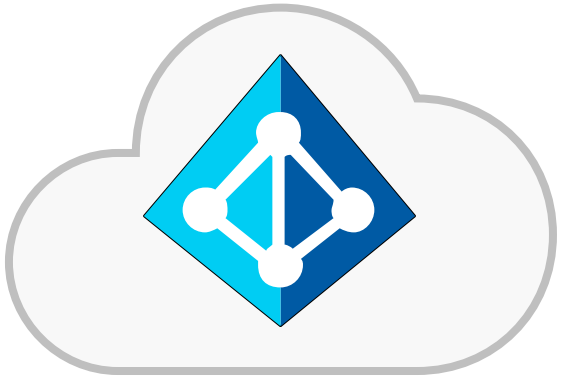
I just wanna log my users in

MSAL.NET makes it simpler ish

- Configure your app(s) in Azure Portal
- Configure your client app
- Profit

My Basic Scenario



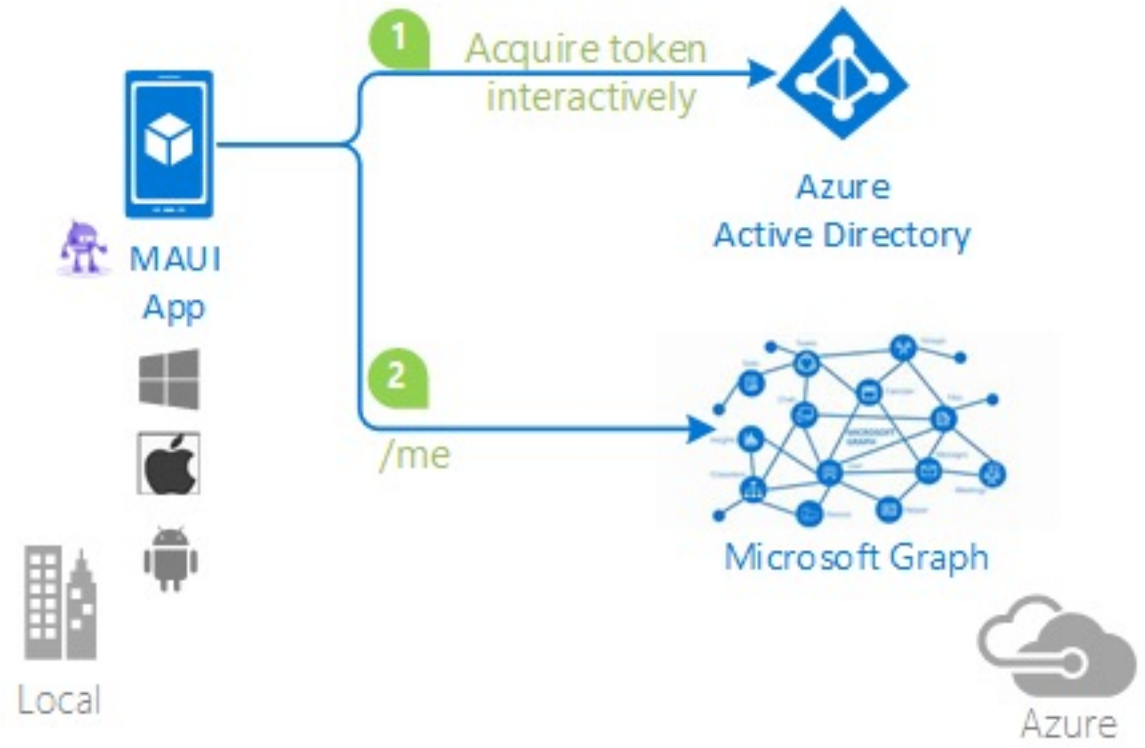


Authorization Required

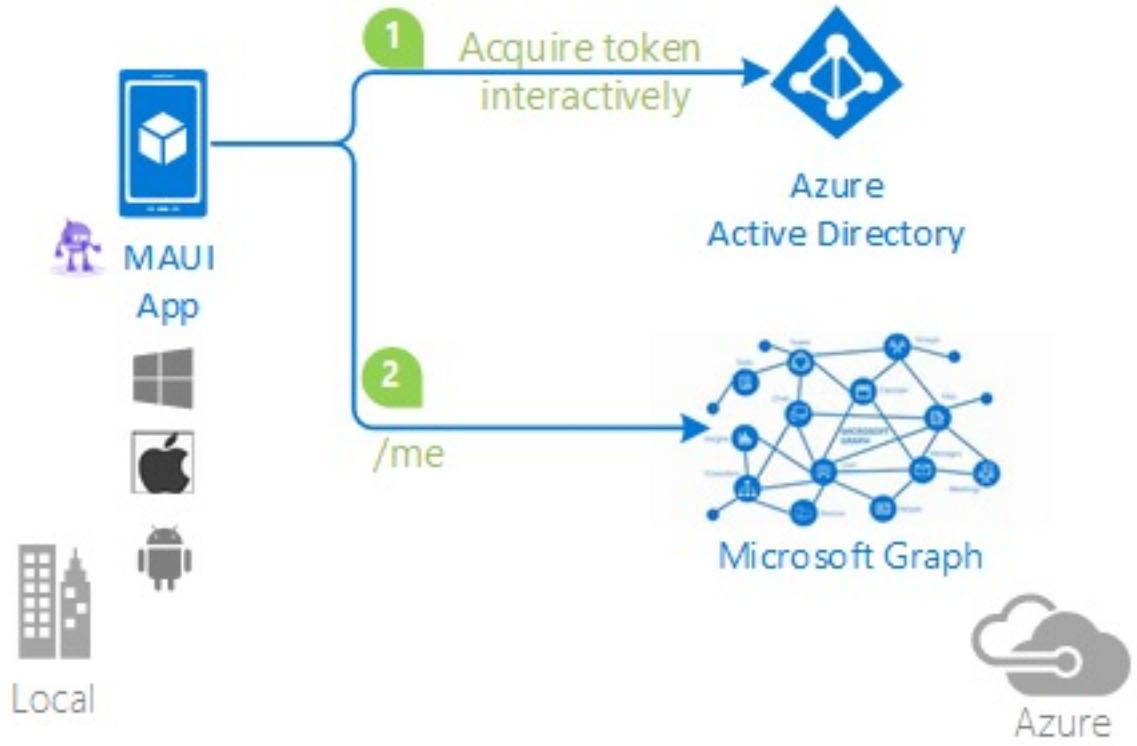


Microsoft Graph User API

Microsoft Identity AzureAD Basic

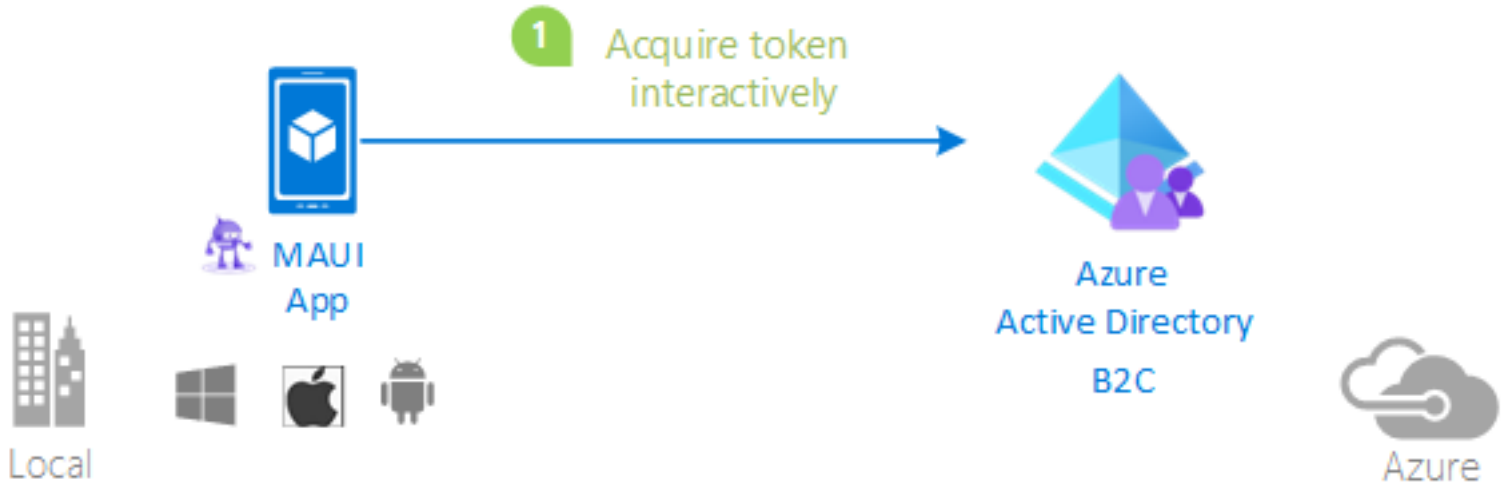


Microsoft Identity
Azure AD Broker



Using a Broker (Microsoft Authenticator) enables more complex scenarios, like device related conditional access and SSO.

Microsoft Identity Azure AD B2C



Identity and access management for your customer-facing apps.

Adds support for third-party platforms such as Google, Facebook, Twitter, LinkedIn, etc.

Resources

- .NET MAUI
- Microsoft.Identity.Client
  - aka MSAL
  - aka MicroSoft Authentication Library
- Microsoft.DataSync.Client
  - aka Azure Mobile Apps
  - aka Microsoft Datasync Framework
- https://github.com/Azure-Samples/ms-identity-dotnetcore-maui
- https://github.com/Azure/azure-mobile-apps



Demos!

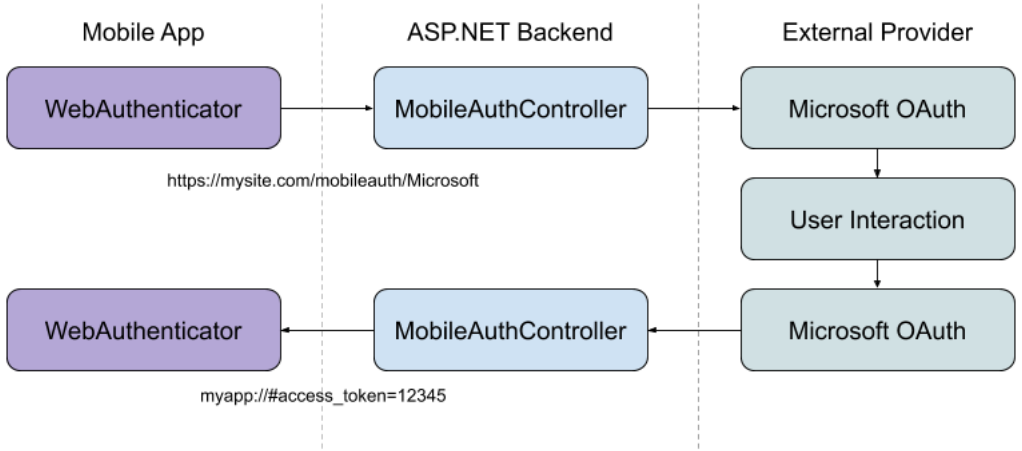
Golden Rules for Tokens

- Use tokens only for their intended function
  - ID Tokens identify the user to the application
  - Access tokens authorize the app for a range of operation for an API
- Never mess with someone else's tokens
  - Apps do not look at access tokens
  - APIs never use ID Tokens for authorization
  - APIs never accept Access Tokens for other APIs
- Cache the tokens appropriately
  - MSAL handles this automatically
- Check for error classes instead of search for specific error codes

Fundamentals of Modern Auth

- Apps request tokens from Microsoft Identity
- Authentication happens over shared web surfaces to provide SSO
- Use the Auth Broker whenever possible
- App passes Access Tokens to APIs
- APIs validate Access Tokens before returning results

WebAuthenticator



The **WebAuthenticator** class lets you initiate browser-based flows which listen for a callback to a specific URL registered to the app.

Microsoft, Facebook, Google, or Apple Sign In account.

https://learn.microsoft.com/dotnet/maui/platform-integration/communication/authentication

Thanks! Let’s talk.

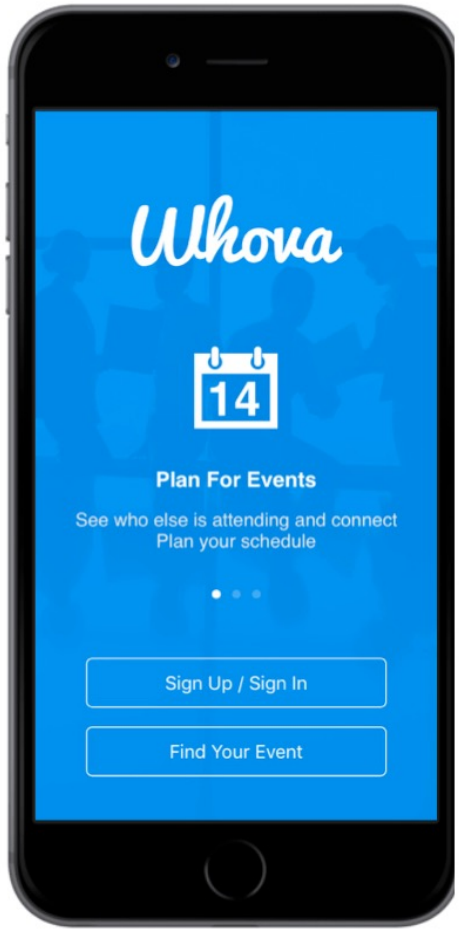
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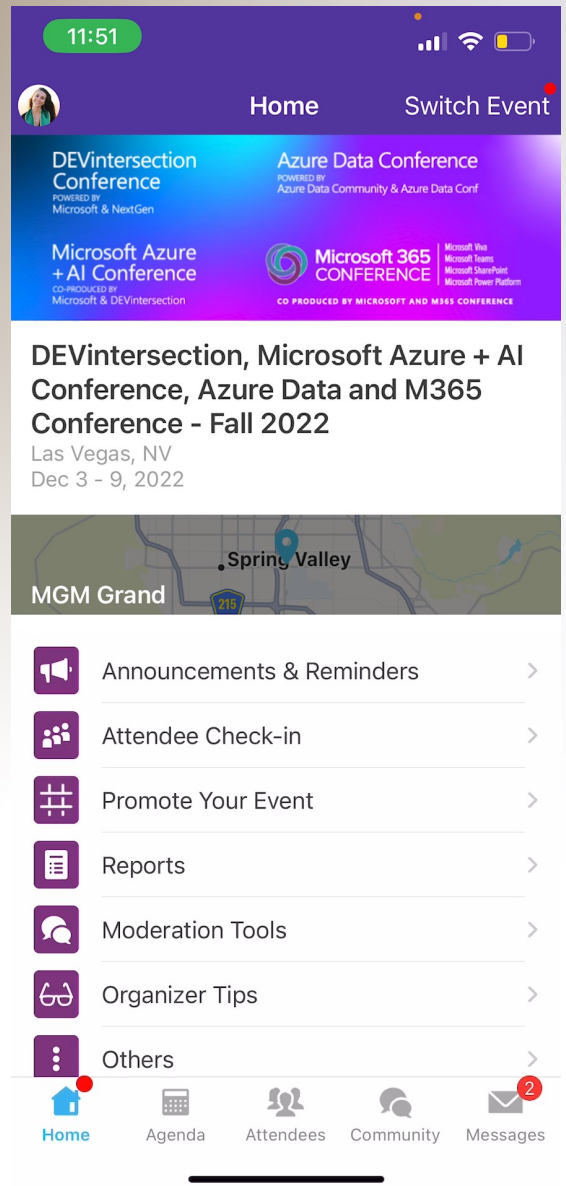


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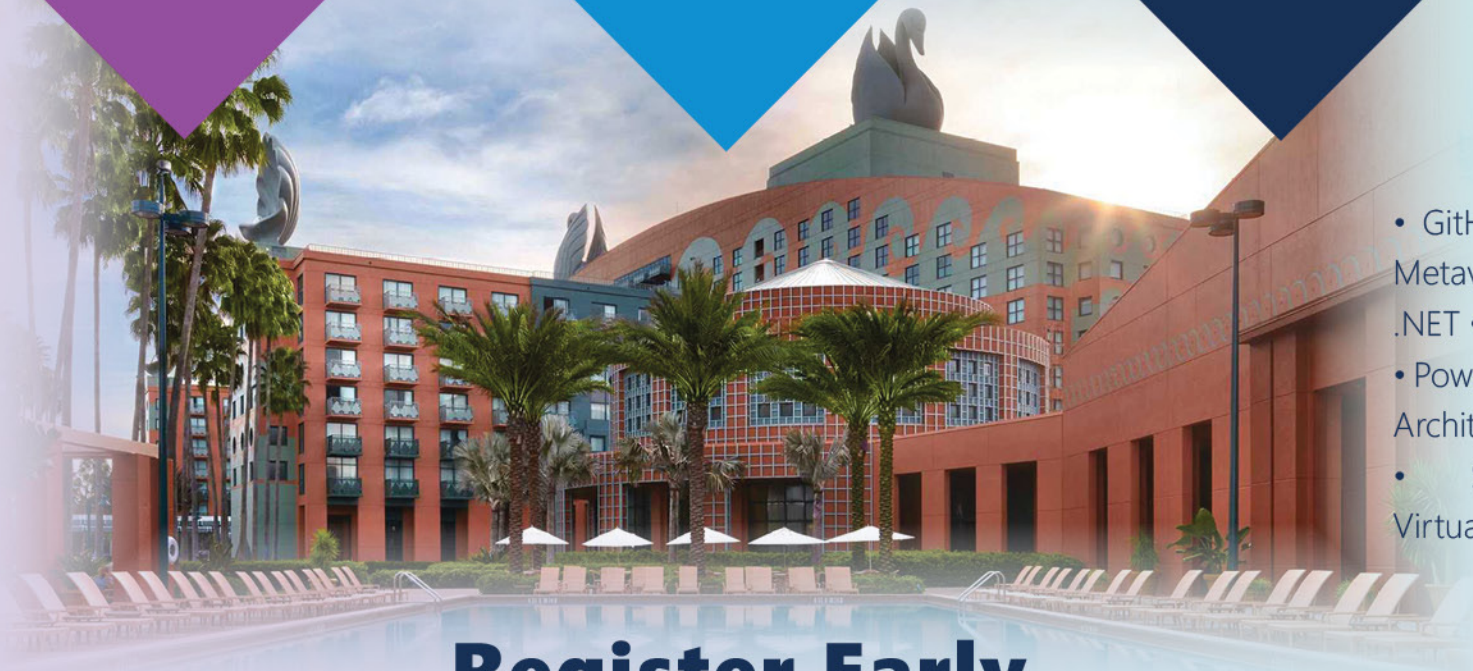
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Other Auth solutions

- Auth0 with Okta https://auth0.com/blog/add-authentication-to-dotnet-maui-apps-with-auth0/